Computational Physics 2 Classes No.6 - Report

Forced oscillations for an electron in a 2D square infinite quantum well

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1 Introduction

In this project the dynamics of an electron confined in a two-dimensional infinite quantum well was investigated using the finite element method (FEM). The system was described by the time-dependent Hamiltonian

$$H(t) = H_0 + eFx \sin \omega t,$$

where H_0 is the stationary Hamiltonian of the infinite quantum well and the second term represents an external oscillating electric field directed along the x-axis.

The time evolution of the wave function was governed by the Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = H(t)\Psi.$$

The wave function was expanded in the FEM basis,

$$\Psi = \sum_{i=1}^{N} c_i(t) h_i(x, y),$$

where the coefficients $c_i(t)$ depend on time. The Crank–Nicolson scheme was applied to obtain the time evolution,

$$\left[\mathbf{S} - \frac{\Delta t}{2\hbar i}\mathbf{H}(t + \Delta t)\right] \mathbf{c}(t + \Delta t) = \left[\mathbf{S} + \frac{\Delta t}{2\hbar i}\mathbf{H}(t)\right] \mathbf{c}(t).$$

The main objective of the simulation was to study resonant transitions between the ground state and the excited states, compare resonant and off-resonant driving, and investigate the leakage of the wave function to higher-energy states.

2 Results

The calculations were performed for a two-dimensional infinite square quantum well using the bilinear finite element basis. The parameters used in the simulation were

$$L = 5, \qquad eF = \frac{1}{L} = 0.2, \qquad \Delta t = 0.25,$$

with $\hbar = 1$.

The generalized eigenvalue problem for the stationary Hamiltonian produced the following lowest positive eigenenergies:

$$E_0 = 0.398041719103,$$

$$E_1 = E_2 = 1.01489206008,$$

$$E_3 = 1.63174240105.$$

The first excited states were found to be degenerate, which is expected for the square infinite well.

The resonant driving frequency was chosen according to

$$\omega = \frac{E_1 - E_0}{\hbar},$$

which gave

$$\omega = 0.616850340975.$$

The off-resonant simulation was performed for

$$\omega = \frac{1}{2}\frac{E_1 - E_0}{\hbar} = 0.308425170488.$$

The norm of the wave function was monitored during the simulation using

$$N(t) = \mathbf{c}^{*T}(t)\mathbf{O}\mathbf{c}(t),$$

and remained close to unity during the entire time evolution, confirming the numerical stability of the Crank–Nicolson propagation scheme.

2.1 Resonant driving

Figure 1 shows the probabilities

$$|p_i(t)|^2$$

for the ground state and the first excited states during resonant driving.

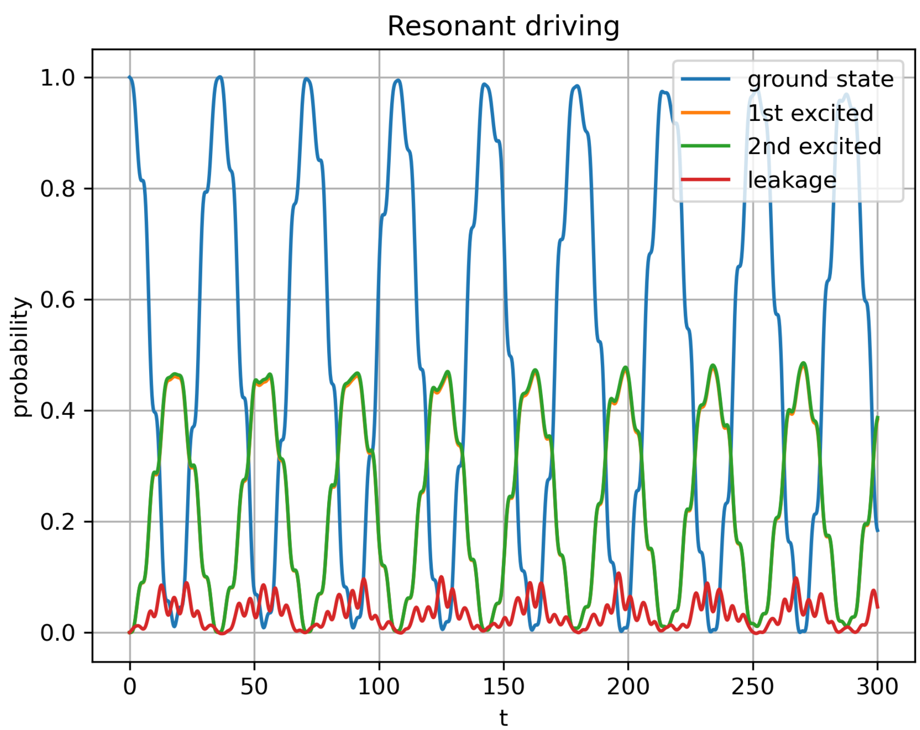

Figure 1: Time evolution of the occupation probabilities for resonant driving.

A strong oscillatory transfer of probability between the ground state and one of the degenerate first excited states can be observed. The probability of the ground-state occupation periodically decreases almost to zero, while the probability of the coupled excited state reaches values close to 0.45.

The second degenerate excited state remains nearly unoccupied throughout the simulation. This behavior is a consequence of the symmetry of the perturbation

$$V(x) = eFx\sin(\omega t),$$

which couples only states with the appropriate parity in the x-direction. Therefore only one of the degenerate first excited states participates significantly in the transition dynamics.

The red curve in Figure 1 represents the leakage to higher-energy states,

$$1 - \sum_{i=0}^{2} |p_i(t)|^2.$$

The leakage remains relatively small during the evolution, typically below 10%, indicating that the dynamics is dominated by transitions between the ground state and a single excited state.

2.2 Off-resonant driving

The time evolution for the off-resonant case, with the driving frequency reduced to one half of the resonant value, is shown in Figure 2.

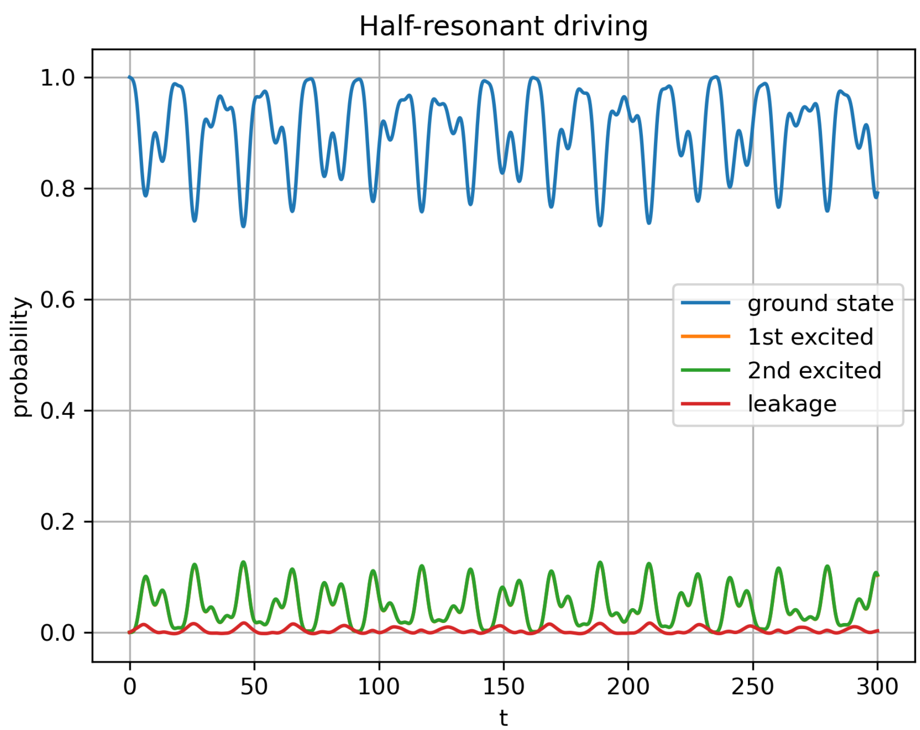

Figure 2: Time evolution of the occupation probabilities for off-resonant driving.

In this case the transition probability is strongly suppressed. The electron remains predominantly in the ground state, whose occupation probability stays close to unity during the entire simulation. Only weak oscillatory excitation of the excited states is observed.

The probability of occupation of the excited states remains small, typically below 0.1, and the leakage to higher-energy states is negligible. This demonstrates the importance of the resonance condition for efficient energy transfer between quantum states.

2.3 Leakage dependence on the electric field amplitude

The leakage to higher-energy states was additionally investigated as a function of the electric field amplitude. The maximum leakage observed during the evolution was measured for several values of eF.

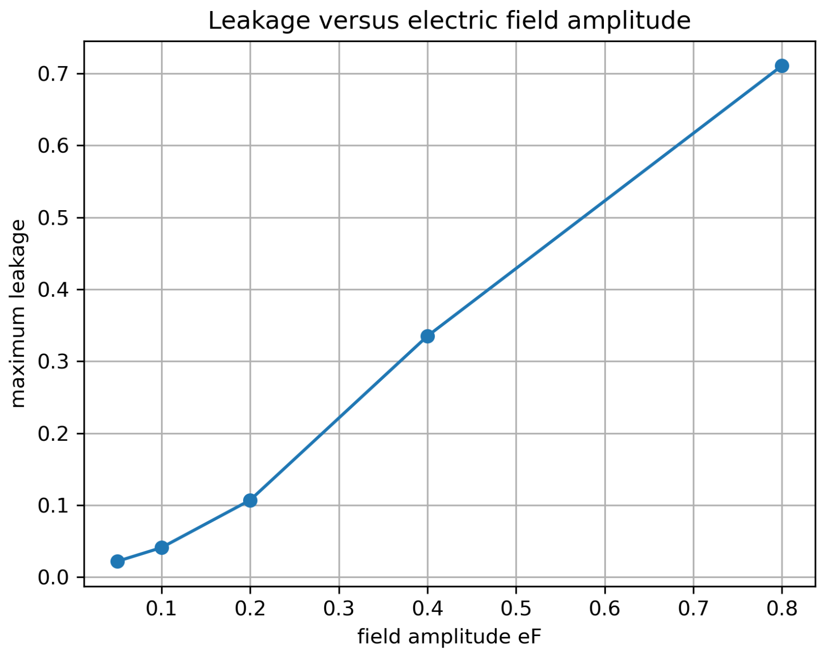

Figure 3: Maximum leakage to higher-energy states as a function of the electric field amplitude.

The obtained values were:

eF	Maximum leakage
0.05	0.0219
0.10	0.0410
0.20	0.1070
0.40	0.3350
0.80	0.7108

The results show that the leakage increases rapidly with the electric field amplitude. For weak driving the dynamics is well approximated by transitions between the ground state and a single excited state. However, for stronger fields the perturbation becomes sufficiently large to populate many higher-energy states, leading to significant leakage and the breakdown of the simple two-level approximation.

3 Conclusions

The forced dynamics of an electron in a two-dimensional infinite quantum well was successfully simulated using the finite element method and the Crank–Nicolson time propagation scheme. Resonant driving produced efficient transitions between the ground state and one of the degenerate first excited states, while off-resonant driving strongly suppressed the transition probability. The simulations also showed that the leakage to higher-energy states increases significantly with the electric field amplitude, demonstrating the breakdown of the simple two-level approximation for strong driving fields.